

Pandas for Machine Learning — Quick Reference

Function / Method	Syntax	Simple explanation
read_csv()	pd.read_csv("file.csv")	Reads a CSV file into a DataFrame.
head()	df.head()	Shows the first 5 rows.
tail()	df.tail()	Shows the last 5 rows.
shape	df.shape	Shows rows and columns count.
columns	df.columns	Shows column names.
dtypes	df.dtypes	Shows data type of each column.
info()	df.info()	Shows column, type, and non-null details.
describe()	df.describe()	Gives a quick statistical summary like count, mean, std, min, max, and quartiles.
loc	df.loc[rows, columns]	Selects rows and columns using labels.
iloc	df.iloc[row, column]	Selects rows and columns using positions.
isna()	df.isna()	Checks for missing values.
isna().sum()	df.isna().sum()	Counts missing values in each column.
fillna()	df["age"].fillna(value)	Fills missing values.
dropna()	df.dropna()	Removes rows with missing values.
duplicated()	df.duplicated()	Checks for duplicate rows.
drop_duplicates()	df.drop_duplicates()	Removes duplicate rows.
value_counts()	df["column"].value_counts()	Counts each value in a column.
unique()	df["column"].unique()	Shows unique values.
nunique()	df["column"].nunique()	Counts unique values.
sort_values()	df.sort_values("column")	Sorts rows by a column.
isin()	df[df["city"].isin(["A", "B"])]	Filters values from a list.
map()	df["gender"].map({"Male":1, "Female":0})	Maps values; useful for simple encoding.
apply()	df["age"].apply(function)	Applies a function; useful for feature engineering.
replace()	df["gender"].replace({"Male":1})	Replaces selected values.
drop()	df.drop("column", axis=1)	Removes a column or row.
rename()	df.rename(columns={"old":"new"})	Changes column names.
mean()	df["age"].mean()	Calculates the average.
median()	df["age"].median()	Finds the middle value.
min()	df["age"].min()	Finds the smallest value.
max()	df["age"].max()	Finds the largest value.
sum()	df["sales"].sum()	Adds values together.
std()	df["age"].std()	Measures data spread.
var()	df["age"].var()	Calculates variance.
groupby()	df.groupby("city")["age"].mean()	Groups data and calculates a summary.
agg()	df.groupby("city")["age"].agg(["min", "max", "mean"])	Calculates multiple statistics.
merge()	pd.merge(df1, df2, on="id")	Combines related DataFrames.
concat()	pd.concat([df1, df2])	Combines DataFrames.
to_datetime()	pd.to_datetime(df["date"])	Converts a column to datetime.
.dt.year	df["date"].dt.year	Extracts the year.
.dt.month	df["date"].dt.month	Extracts the month.
.dt.day	df["date"].dt.day	Extracts the day.
reset_index()	df.reset_index(drop=True)	Resets the DataFrame index.
pivot_table()	pd.pivot_table(df, values="sales", index="category", aggfunc="sum")	Creates a summary table.
to_csv()	df.to_csv("cleaned.csv", index=False)	Saves a DataFrame as CSV.
X (features)	X = df.drop("target", axis=1)	Creates ML input features.
y (target)	y = df["target"]	Creates the ML target/output.